Spacer composition, not motif arrangement, dominates apparent regulatory grammar detected by DNA foundation models

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Motivation

Whether the spatial arrangement of transcription factor (TF) binding sites encodes functional information—a "regulatory grammar"—or whether enhancers operate as "billboards" where only TF identity matters remains debated [1,2]. DNA foundation models enable systematic *in silico* perturbation of enhancer architecture, but standard grammar sensitivity metrics confound motif arrangement with spacer DNA composition. We introduce a **spacer-factored grammar sensitivity framework (SF-GSI)** applied across four foundation models, five MPRA datasets, and four organisms to quantify the true contribution of motif arrangement to gene expression.

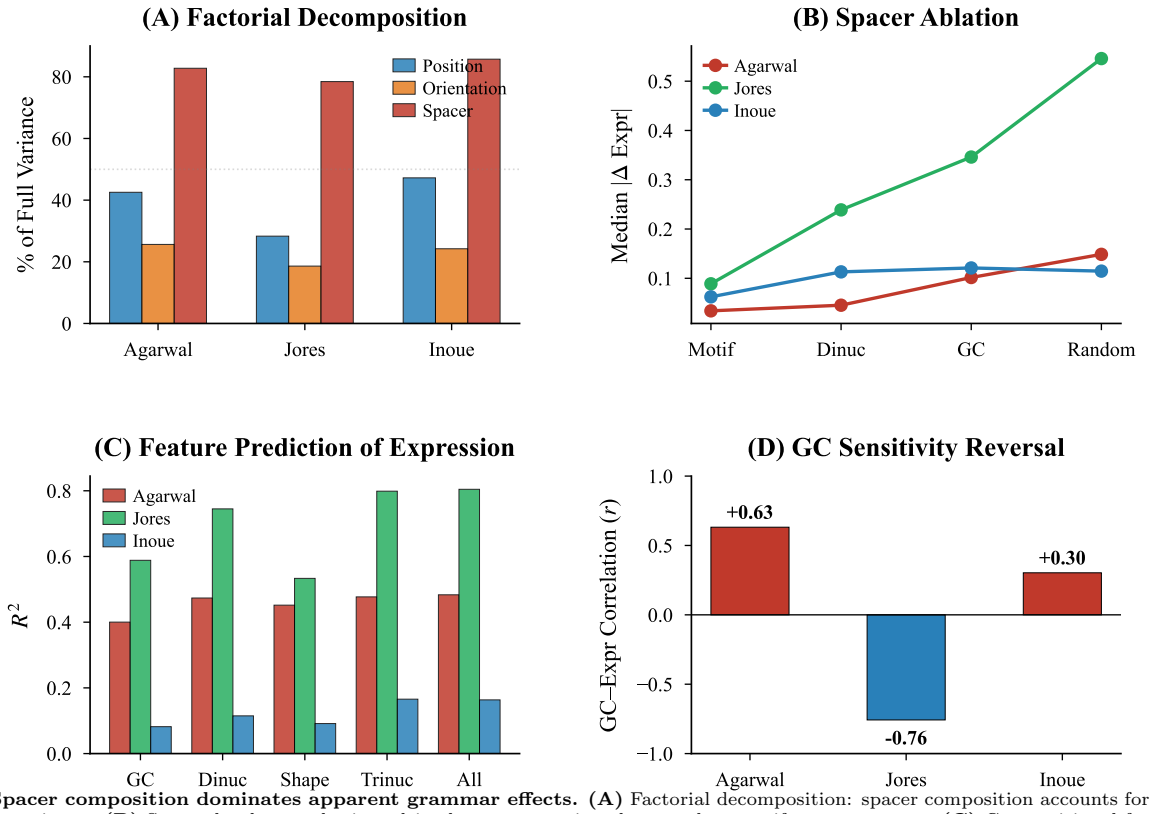

Figure 1: Spacer composition dominates apparent grammar effects. (A) Factorial decomposition: spacer composition accounts for the majority of expression variance. **(B)** Spacer-level perturbations drive larger expression changes than motif rearrangement. **(C)** Compositional features predict 40–80% of model R^2. **(D)** GC–expression correlation reverses across species ($r = +0.63$ human vs. $r = -0.76$ plant).

Methods

Grammar Sensitivity Index (GSI). TF motifs are identified via FIMO against JASPAR 2024 [3]; 1,000 vocabulary-preserving shuffles permute motif order while holding identity constant. GSI $= \sigma(\hat{y})/|\mu(\hat{y})|$. **Spacer-Factored GSI (SF-GSI).** A factorial design isolates: (i) motif position/orientation (true grammar), (ii) spacer composition (confound), (iii) flanking-context effects via matched-spacer shuffles and regression residuals. **Rule Extraction.** Pairwise rules via spacing scans (2–50 bp), four orientation configurations, and helical phasing at ∼10.5 bp. **Variance Decomposition.** Hierarchical ANOVA: vocabulary → +grammar → +higher-order → full model. **Models & Data.** DNABERT-2, Nucleotide Transformer v2-500M (transformers), HyenaDNA (SSM), PARM (CNN); five MPRA datasets across human, yeast, and plant (∼7,650 enhancers) [4,5].

Results

The spacer confound. Spacer DNA composition accounts for 78–86% of measured GSI variance (Fig. 1A–B). Simple compositional statistics explain 40–80% of predictions (Fig. 1C), and the GC–expression relationship reverses between species (Fig. 1D).

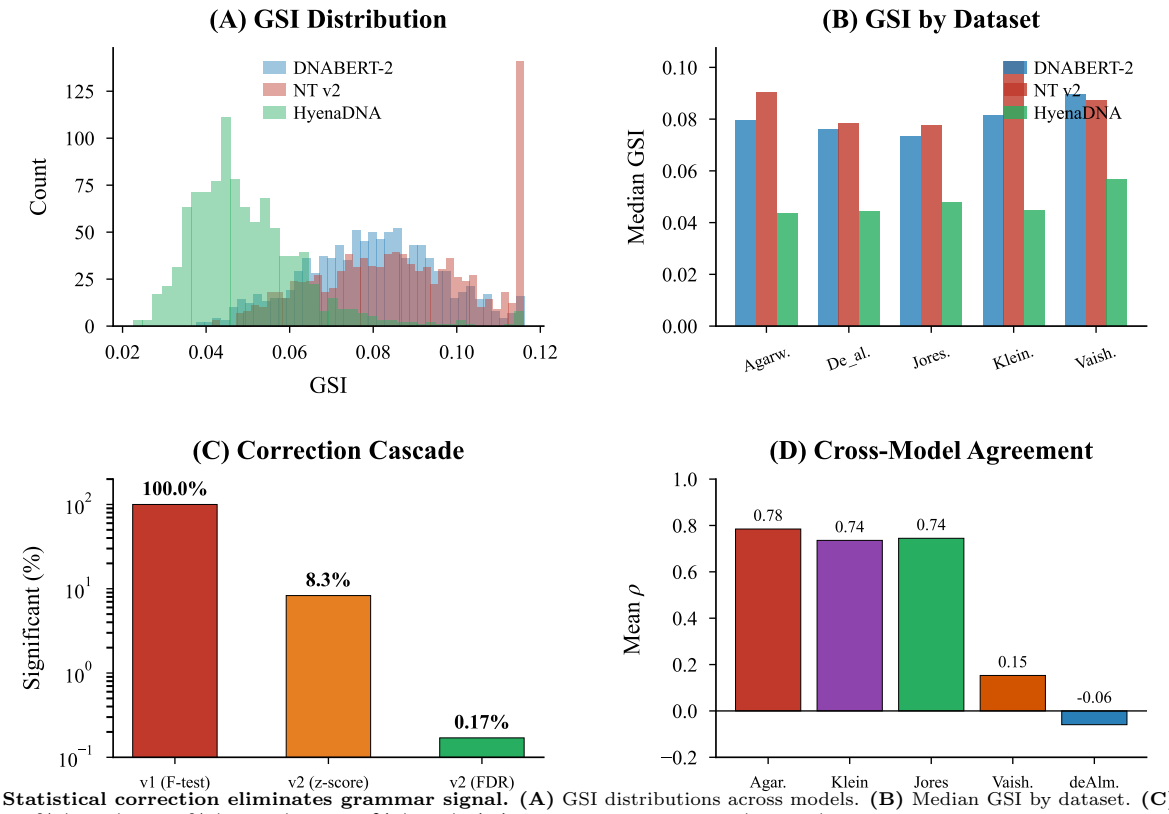

Figure 2: Statistical correction eliminates grammar signal. (A) GSI distributions across models. **(B)** Median GSI by dataset. **(C)** Significant enhancers: 100% (naive) → 8.3% (z-score) → **0.17%** (FDR). **(D)** Cross-model agreement ($\rho \approx 0.74$).

Correction cascade. Naive testing yields 100% "grammatical" enhancers; z-score normalization reduces this to 8.3%, and FDR correction to **0.17%**—a three-order-of-magnitude drop consistent across all architectures (Fig. 2C–D).

Positive controls. With spacer composition held constant, orientation effects are detected at $p < 10^{-117}$ with 100% detection rate (vs. 8.3% under standard shuffles), confirming methodological sensitivity; the null result reflects biology.

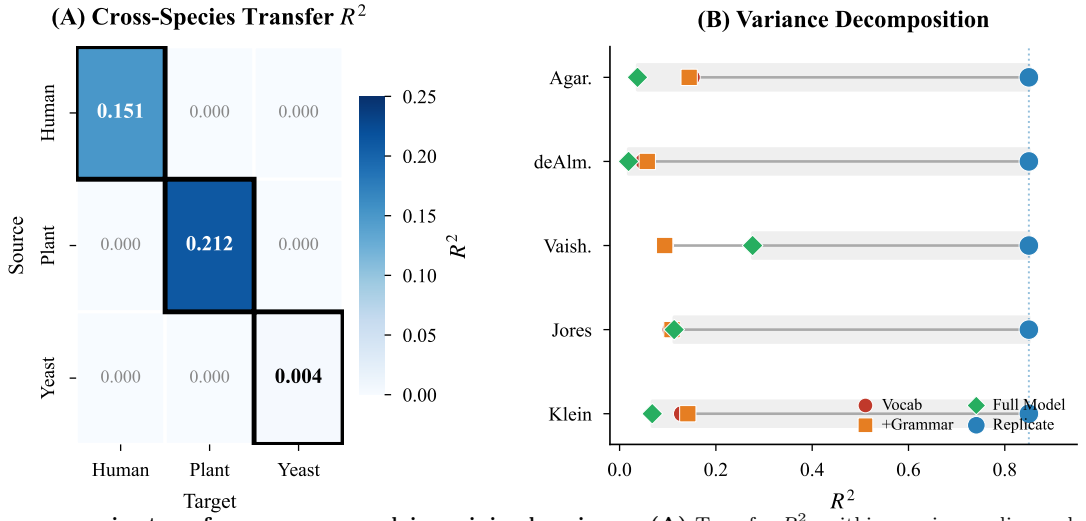

Figure 3: No cross-species transfer; grammar explains minimal variance. (A) Transfer R^2: within-species on diagonal; off-diagonal ≈ 0. **(B)** Vocab + grammar <22% R^2; gap to replicate ceiling (0.85) is non-sequence.

Transfer & variance decomposition. Grammar rules from one organism predict near-zero effects in others (Fig. 3A). Motif identity explains 8–22% of variance; grammar adds only 0–1.6%. The gap to replicate ceilings (∼85%) indicates most variance is attributable to *trans*-regulatory factors (Fig. 3B).

Conclusion

Regulatory DNA operates predominantly as a *billboard*: TF identity, not arrangement, determines enhancer activity. Apparent grammar is largely spacer composition—a confound inflating sensitivity ∼1,000×. Findings hold across four architectures, five datasets, and four organisms. SF-GSI provides the statistical machinery to separate grammatical effects from compositional confounds in regulatory genomics. Code: `https://github.com/bryanc5864/GRAMLANG`.

References: [1] Arnosti & Kulkarni, *J. Cell. Biochem.* 94:890, 2005. [2] Long et al., *Cell* 167:1170, 2016. [3] Castro-Mondragon et al., *Nucleic Acids Res.* 52:D142, 2024. [4] Agarwal et al., *Science* 380:eadh0559, 2023. [5] Jores et al., *Nat. Plants* 7:842, 2021.